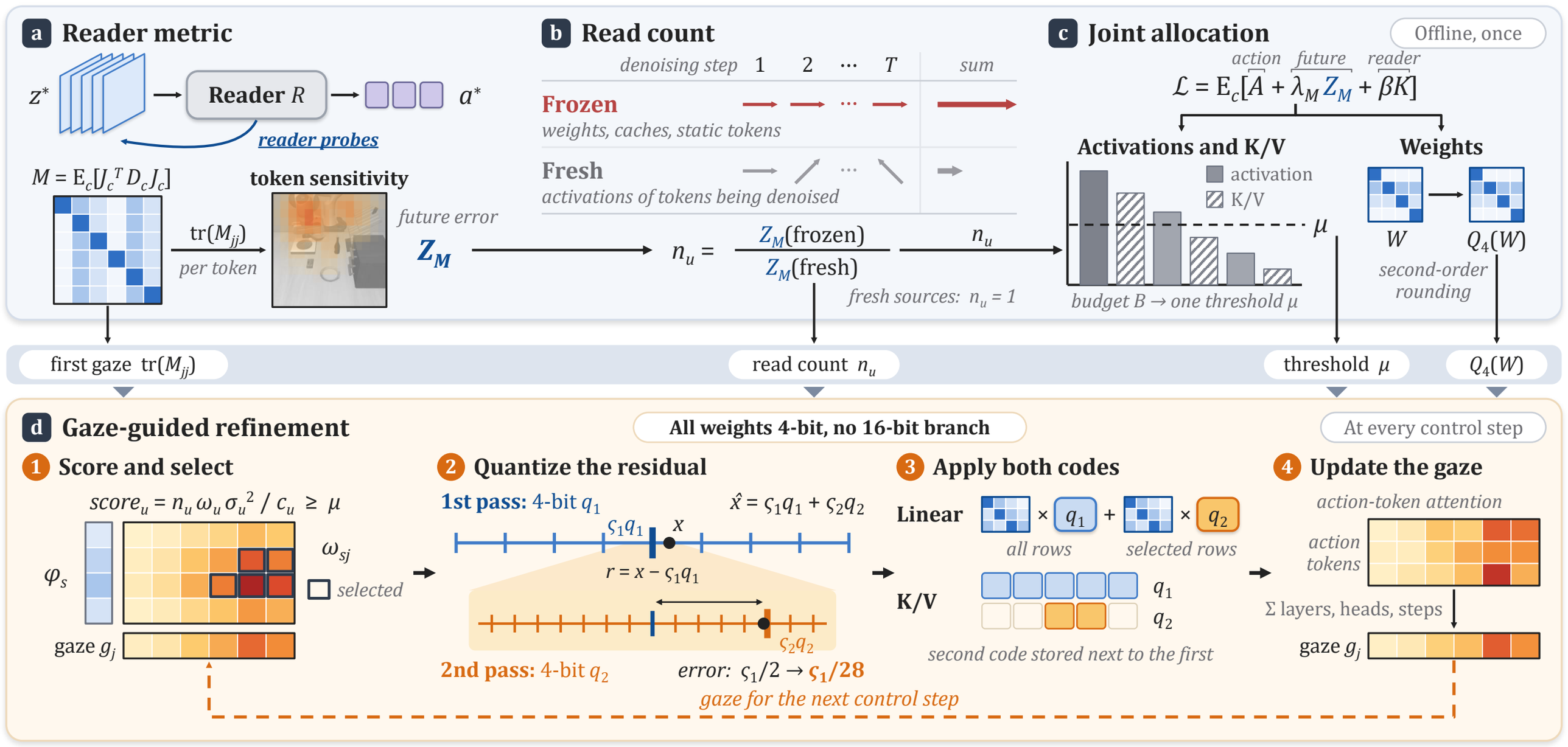

a Reader metric
Reader R
reader probes
token sensitivity
future error
per token
b Read count
denoising step 1 2 ⋯ T sum
Frozen
weights, caches, static tokens
Fresh
activations of tokens being denoised
fresh sources: n_u = 1
c Joint allocation
Offline, once
action future reader
Activations and K/V
Weights
activation
K/V
budget B → one threshold μ
second-order rounding
first gaze tr(M_jj)
read count n_u
threshold μ
Q_4(W)
d Gaze-guided refinement
All weights 4-bit, no 16-bit branch
At every control step
1 Score and select
selected
gaze g_j
2 Quantize the residual
1st pass: 4-bit q_1
2nd pass: 4-bit q_2
error: ς_1/2 → ς_1/28
3 Apply both codes
Linear
all rows
selected rows
K/V
second code stored next to the first
4 Update the gaze
action-token attention
action tokens
Σ layers, heads, steps
gaze for the next control step